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Nickel foam–caged Ag-Au bimetallic nanostructure as a highly rugged and durable SERS substrate

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ABSTRACT

A three-dimensional nickel foam (NF) caging Ag-Au bimetallic nanostructure (Ag-Au@NF) has been demonstrated as a highly rugged and sensitive surface-enhanced Raman scattering (SERS) substrate. The main concept employed in designing this substrate was the protection of SERS-active nanostructures by holding them within a sturdy frame. For this purpose, NF was readily chosen as a strong porous frame to secure the internal nanostructures owing to its excellent structural strength and ability to easily incorporate analytes due to its porous structure. The next issue was the construction of stable and SERS-efficient nanostructures within the NF frame. To meet this need, an Ag-Au bimetallic nanostructure was chosen, combining the high SERS efficiency of Ag and the good surface stability of Au. To fabricate the Ag-Au nanostructure, first a Sn-Ag alloy was constructed on the NF frame by means of electrodeposition (producing Sn-Ag@NF) and then Sn was selectively leached from the alloy by means of free corrosion in NaOH solution (producing Ag@NF). This selective leaching was adopted to make the surface rough to enhance the SERS efficiency in the final substrate. Finally, Au was incorporated into the Ag@NF by means of a galvanic replacement reaction (producing Ag-Au@NF). The use of Ag-Au@NF enabled identification of rhodamine 6G at concentrations as low as 0.1 nM. The Ag-Au@NF substrates were rugged, showing small Raman signal drops after 100 cycles of sandpaper abrasion or 30 min of sonication. Also, the signal decrease was only 4.26% upon 3 days of exposure to ambient air without special care. Overall, the Ag-Au@NF developed herein shows practical merits for adoption in routine and field SERS analyses.

1. Introduction

Surface enhanced Raman scattering (SERS) has been increasingly adopted for analysis of diverse samples [1–5] because of its high sensitivity, involving the enhancement of a molecule's Raman signal by resonance with a local plasmon generated on a SERS substrate. Because the substrate used is a key factor governing the sensitivity and reproducibility of a given SERS analysis, many different SERS substrates composed of noble metals (Ag, Au, and Cu) have been developed such as nanoparticles [6], nanostars [7], nanorods [8,9], and nanodendrites [10]. SERS-based analysis tends to require a well-controlled analytical environment because SERS-generating nanostructures would be easily destroyed by physical contact and vibration. Therefore, to extend the practical applications of SERS to more challenging environments, for example at-site field analysis using a portable Raman system, it is necessary to increase the ruggedness and chemical stability of the SERS substrate.

One versatile approach to meet these requirements is to use SERS-active nanostructures protected inside bulk porous materials such as silver-grafted sponge [11] and melamine sponge [12], or water-uptaking polymers such as a hydrogel [13] and agarose gel [14]. The use of porous material or water-swellable polymer allows analytes to access the interior SERS-active nanostructures. Use of this approach protects the active materials from incidental or accidental physical contact, and the porous substrate is also advantageous for facile introduction of analytes inside. In a meanwhile, the housing materials in these researches are rather soft and flexible, so it is desirable to develop more durable and sturdy SERS substrates to ensure the ruggedness critically demanded for routine and field analysis; such as high physical endurance and ease of use by non-specialists.

Nickel foam (NF) [15–17] is a three-dimensional (3D) material that is an interesting candidate as a strong frame owing to its high durability. Similar to sponge, it is highly porous (with 75–95% void volume) and its physical strength is superior. A few research groups have built SERS substrates inside the porous structure of NF, for example Ag nanoparticle films [18], Au nanoparticles [19], and graphene foam decorated with Ag nanoparticles [20], and have demonstrated their utility in SERS-based detection of melamine (10 μM), pyrene (10 nM),

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and methylene blue (1 nM), respectively. Although these studies have included the use of monometallic SERS-active nanomaterials, multimetallic SERS-active materials could be used to combine the individual merits of the different metals in a single structure and thereby improve SERS efficiency [21,22].

In the present work, to make a highly rugged and efficient SERS substrate, we constructed Ag-Au bimetallic nanostructures in a NF frame (hereinafter abbreviated as Ag-Au@NF). The choice of this dual-metal composition was to ensure sensitive and stable SERS analysis by simultaneously taking advantage of the high SERS activity of Ag and the high surface stability of Au, and NF was adopted as a strong frame to protect the Ag-Au nanostructures inside and thereby to increase the overall ruggedness of the substrate. To build the Ag-Au nanostructures, a Sn-Ag alloy was initially built within the NF framework (abbreviated Sn-Ag@NF) by means of electrochemical co-deposition in a mixture solution of $AgNO_3$ and $SnSO_4$. Then, the Sn-Ag@NF was immersed in NaOH solution to selectively leach the Sn, yielding Ag@NF. This two-step procedure was applied rather than direct deposition of Ag to disperse Ag homogeneously throughout the porous structure by means of competitive deposition with Sn and to generate a rough surface after the selective leaching of Sn. Finally, Au was incorporated into the Ag@NF by means of galvanic replacement reaction (GRR) to yield Ag-Au@NF.

The structures of the synthesized Sn-Ag@NF, Ag@NF, and Ag-Au@NF were characterized using scanning electron microscopy (SEM), and the surface of Ag-Au@NF was further characterized by means of atomic force microscopy (AFM). Using the Ag-Au@NF substrate, SERS spectra of rhodamine 6G (R6G) samples ranging in concentration from 0.1 nM to 1.0 μM were acquired, and the sensitivity of the substrate was assessed based upon the results. To evaluate the durability of Ag-Au@NF substrates, the variations in SERS signals of 100 μM R6G samples were evaluated under two separate harsh treatments: 100 cycles of sandpaper abrasion and sonication for 30 min. Finally, to assess long-term surface stability, spectra of R6G adsorbed on Ag-Au@NF were collected over a period of 25 days and the resulting intensity variation was analyzed.

2. Material and methods

2.1. Chemicals and materials

$AgNO_3$ (99.8 wt%, Junsei Chemical, Japan) and $SnSO_4$ (95%, Junsei Chemical) were used to prepare 0.1 M Sn and 0.01 M Ag electroplating solutions, respectively. $HAuCl_4$ (99.9 wt%, Sigma-Aldrich) was used to prepare a 5.0 mM Au solution for galvanic displacement. Other chemicals used in this study, H_2SO_4 (J.T Baker), NaOH (Sigma-Aldrich), and acetone (Daejung, Korea), were of analytical grade. Nickel foam (NF) was purchased from MTI-Korea (Seoul, Korea). A 1.0 mM R6G (Sigma-Aldrich) stock solution was prepared using doubly distilled water (DDW). R6G samples of various concentrations were prepared by diluting the stock solution.

2.2. Fabrication of Ag-Au@NF

NF (0.8 cm × 1.5 cm) was initially cleaned by sonication with acetone, and then further rinsed in sulfuric acid (0.5 M) and DDW to eliminate possible organic and inorganic contaminants on NF. To synthesize Sn-Ag@NF, the cleaned NF was directly used as a working electrode without any additional solid electrode supports and was immersed in 4 mL of a solution 0.01 M in $AgNO_3$, 0.1 M in $SnSO_4$, and 0.5 M in H_2SO_4 for electrochemical deposition. The electrodeposition was conducted by applying current of −5 mA to the NF using a galvanostat (EDAQ, Australia) over 5 min with stirring, using a Pt counter electrode and a Ag/AgCl reference electrode. The produced Sn-Ag@NF was carefully washed with DDW.

Sn was selectively leached from Sn-Ag@NF by immersing the substrate in a 5.0 M NaOH solution for 24 h. Then, the remaining Ag@NF structure was carefully rinsed with DDW several times and dried in air. Finally, Au was incorporated into the Ag@NF by means of spontaneous GRR in a 5.0 mM $HAuCl_4$ solution, yielding Ag-Au@NF. The length of GRR was varied from 5 to 50 s to find an optimal condition providing maximum SERS intensity.

2.3. Characterization of Ag-Au@NF structure and SERS measurement

SEM images and energy dispersive X-ray (EDX) spectra of built substrates were acquired using a Hitachi S-4800 SEM instrument. Transmission electron microscopy (TEM) and atomic mapping images were acquired using a JEM-2100 F TEM instrument (JEOL, Japan). AFM images were acquired using a XE-100 microscope (Park System Corp., Suwon, Korea) equipped with 910M-NCHR silicon cantilevers and operated in a non-contact mode. All Raman spectra of R6G-adsorbed Ag-Au@NF substrates were collected in air using a wide area coverage (WAC) scheme, with the laser (785 nm) illumination diameter of 6.0 mm (area: 28.3 mm^2; PhAT probe, Kaiser Optical Systems). Each substrate was incubated in a R6G sample solution over 5 min for the adsorption of R6G (via the contained amine group) on the substrate surface. Then, the R6G-adsorbed substrate was carefully rinsed with distilled water to wash out unbound R6G molecules and naturally dried in air over approximately 50 min without additional purging. For each test, such as R6G measurement and durability evaluation, three substrates were separately prepared and Raman spectra were acquired at three random locations on each substrate with the exposure time of 10 s. Baseline correction of spectra was performed using Matlab version 7.0 (The MathWorks, Inc., MA, USA).

3. Results and discussion

3.1. Examination of Ag-Au@NF structure and SERS measurement

A SEM image of bare NF clearly showed the frame and confirmed the 3D porous structure (Fig. 1(a)). The pore size was approximately 250 μm. Initially, Sn-Ag alloy was incorporated into the NF frame by means of the electrochemical co-deposition of Sn and Ag described earlier. A SEM image of the Sn-Ag alloy formed on the NF frame shows its smooth surface with white spots somewhat broadly spread over the image (Fig. 1(b)). EDX analysis reveals that the structure contains both Sn and Ag (refer to Figure S1 (a) in the supplementary data). A SEM image of the Ag@NF prepared by dealloying of Sn-Ag@NF shows its considerably roughened surface (Fig. 1(c)). The roughened surface is advantageous in increasing surface area and increasing the chance of forming SERS hotspots. EDX analysis confirmed the absence of Sn in the Ag@NF (Figure S1 (b) in the supplementary data).

One important issue to consider in acquiring the spectra was the potential SERS intensity fluctuation depending on the location of laser illumination on the porous substrate. Because the final Ag-Au@NF material has large pores (~250 μm), the SERS intensity should fluctuate as the laser passes over the frame and void spaces. One simple and effective approach to minimize this problem is to use large-area sampling. Therefore, a WAC scheme with the illumination area of 28.3 mm^2 (diameter: 6 mm) was employed throughout this study. This canceled the intensity fluctuations, thereby realizing reproducible spectral acquisition as demonstrated previously [10,23].

To optimize the Ag-Au@NF to yield the greatest SERS intensity, the GRR duration was varied from 5 to 50 s and the resulting substrates were used to measure 100 μM R6G. The Ag-Au@NF prepared with GRR of 20 s provided the highest Raman peak intensity among the substrates tested (Fig. 2). EDX analysis confirmed the presence of Au and Ag (refer to Figure S1c in the supplementary data). SEM images of this 20 s GRR formulation are shown with three different magnifications in Fig. 1(d) through (f). After the incorporation of Au, the surface morphology became mostly homogeneous and showed closely packed nanoclusters.

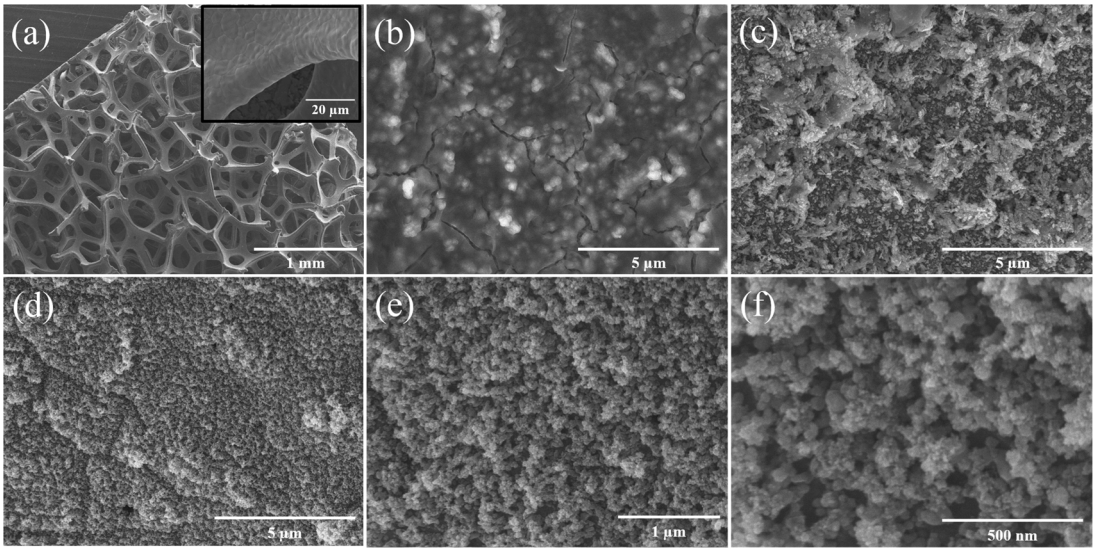

Fig. 1. SEM images of (a) NF, (b) Sn-Ag@NF, and (c) Ag@NF.. SEM images of Ag-Au@NF with three different magnifications ((d) through (f)).

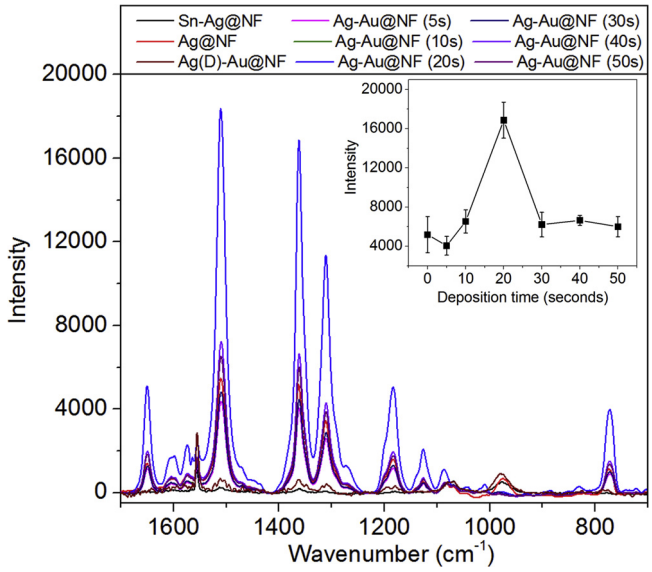

Fig. 2. Raman spectra of 100 µM R6G acquired using Sn-Ag@NF, Ag@NF, Ag (D)-Au@NF substrates and Ag-Au@NF substrates prepared by means of GRR of various durations (5, 10, 20, 30, 40, and 50 s). (Inset) Raman intensity at 1360 cm^{-1} versus GRR duration applied.

These dense Ag-Au bimetallic nanoclusters were responsible for the enhanced SERS intensity. The optimal Ag-Au@NF structure gave 3.25 times higher Raman intensity compared to that provided by the Ag@NF structure. The Ag-Au bimetallic composition is effective in enhancing SERS efficiency [21,22]. For deep examination, Raman mapping was performed on R6G-adsorbed Ag-Au@NF (concentration: 100 µM) performed as described in the supplementary data (Figure S2). As shown, the red color (corresponding to high R6G peak intensity at 1360 cm^{-1}) was clearly observed throughout the frame, thereby indicating the homogeneous distribution of Ag-Au structure on the frame. Further, enhancement factor (EF) of the substrate was calculated and the achieved EF was 2.46×10^8. EFs reported in relevant previous five publications [24–28], in which diverse Ag-Au bimetallic nanostructures were employed and R6G was used as a Raman dye, ranged from 9.0×10^4 to 1.0×10^{10}. In comparison with the EFs in the previous reports, the EF achieved Ag-Au@NF using is considerably high and comparable. In a meanwhile, when Sn-Ag@NF was used for the same measurement, (Fig. 2). This occurred because Sn covered most of the Ag and blocked access of the analyte to the Ag surface to allow generation of a SERS signal.

TEM and elemental analysis were conducted on a piece of Ag-Au bimetal detached from the Ag-Au@NF (Fig. 3), showing both metals were present homogeneously throughout the sample without noticeable localization. The atomic percentages of Ag and Au were determined to be 44.1% and 55.9%, respectively; thus, the surface can be expected to be more stable than that of the Ag-only structure due to the incorporation of Au. In addition, the XRD spectrum of Ag-Au@NF as well as UV/Vis spectrum of Ag-Au structure are provided in the supplementary data (Figure S3 and S4). As shown, the built Ag-Au structure on NF was crystalline and its plasmonic bands at 410 and 535 nm were apparent with the broad absorption over the visible spectral range (Figure S4).

The surface topography of the Ag-Au nanostructure was further characterized by AFM (Fig. 4). It has a fairly regular dimpling pattern

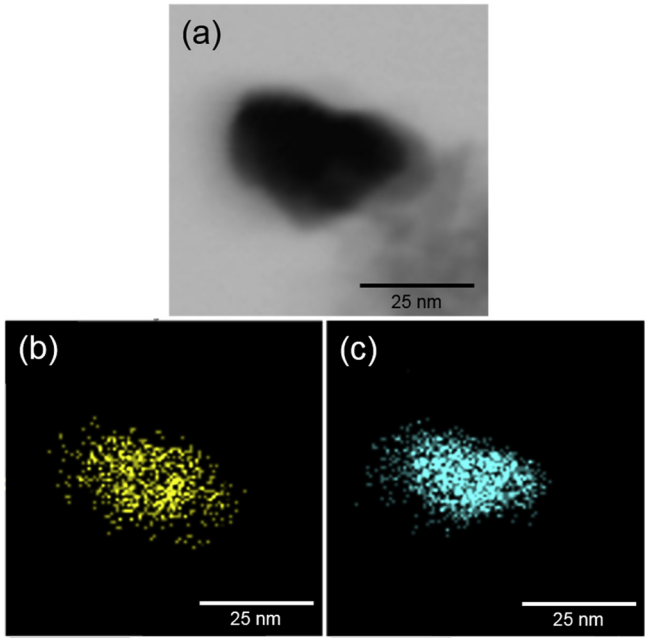

Fig. 3. TEM image of a piece of Ag-Au bimetal detached from the Ag-Au@NF (a) and the corresponding elemental maps of Ag (b) and Au (c).

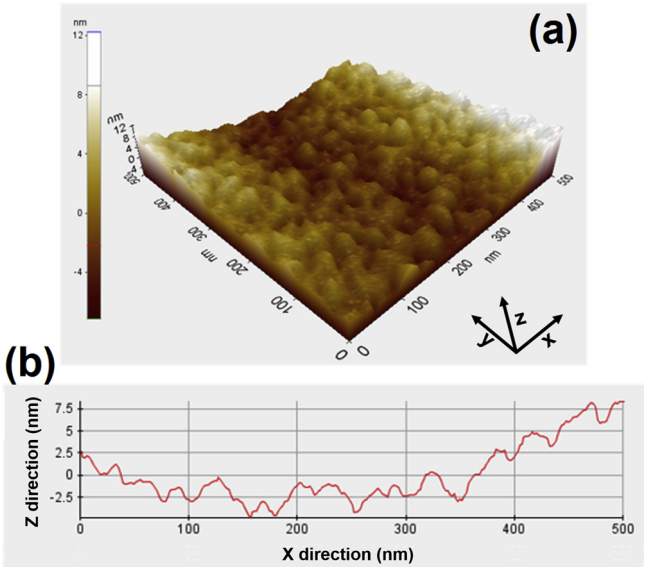

Fig. 4. Three-dimensional (3D) topography of a 500 × 500 nm^2 area of Ag-Au@NF (a); *x*-direction line topology of Ag-Au@NF acquired using AFM (b).

with homogeneous morphology. The calculated root mean square surface roughness was 3.54 nm and the average nanostructure size was approximately 30 nm. This nanoscale surface roughness was a main attribute promoting the strong SERS intensity observed for R6G (refer to Fig. 2).

As mentioned above, it is possible to directly deposit Ag onto the NF rather than using the two-step process of Sn-Ag alloy deposition and subsequent leaching of Sn, although the latter process is expected to provide greater SERS activity. To demonstrate the difference between these two cases, a substrate was constructed by means of direct deposition of Ag onto NF; the product is herein denoted as Ag(D)@NF to distinguish it from Ag@NF. In Ag(D)@NF, Ag thickly covered the NF pores, whereas the NF pores and frame were clearly observable in Ag@NF (Fig. 5). Because Ag had no competition with other metals during its deposition to form Ag(D)@NF, Ag easily grew to a bulk agglomerate. Moreover, the Ag agglomerate in Ag(D)@NF was loosely bound and washed away easily from the NF, even during mild washing with water. Incorporating Au into this structure yielded a substrate (denoted as Ag(D)-Au@NF) with very little SERS activity for 100 μM R6 G (Fig. 2). This demonstrated the advantage of using the two-step process to build a more efficient SERS substrate.

To evaluate the sensitivity, Raman spectra of five R6G samples in a low concentration range (0.1 nM to 1.0 μM) were collected using Ag-Au@NF. As described above, three separately prepared substrates were

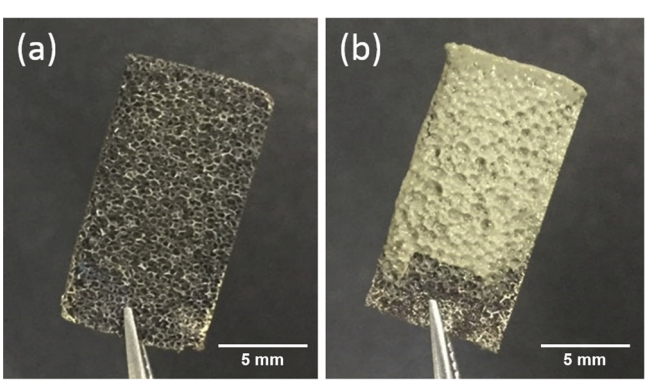

Fig. 5. Photographs of (a) Ag@NF and (b) Ag(D)@NF, respectively prepared by means of a two-step process (electrodeposition of Sn-Ag alloy followed by selective leaching of Sn) and one-step direct Ag deposition onto NF.

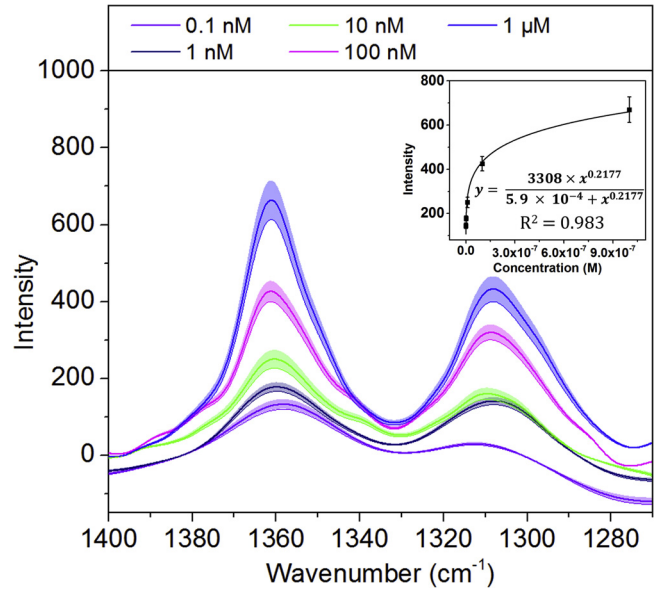

Fig. 6. Raman spectra of five R6 G samples in a low concentration range (0.1 nM to 1 μM), acquired using Ag-Au@NF. Shades overlapping each spectrum indicate standard deviations based on 9 observations (3 separately prepared substrates and 3 measurements for each case). (Inset) Raman intensity at 1360 cm^{-1} versus R6G concentration.

used for each sample measurement and three spectra were acquired for each by illuminating the wide-area laser at three different locations. Fig. 6 shows the acquired spectra in the 1400–1270 cm^{-1} range, where two major R6G peaks appear. The shades overlapping each spectrum indicate the corresponding standard deviations based on the nine observations. The inset indicates the peak intensity at 1360 cm^{-1} versus the R6G concentration. The intensity variation followed the Langmuir isotherm model [29], as also observed in other SERS measurements. The fitting formula and the resulting R^2 are designated in the inset. The average relative standard deviation (RSD) of the measurement was 8.12%, indicating good substrate-to-substrate reproducibility. The achieved reproducibility was mainly attributed to the use of WAC scheme able to compensate for fluctuation of Raman intensities of R6Gs adsorbed at different locations especially on the substrate containing large pores. In the spectrum for the 0.1 nM R6G sample, the 1360 cm^{-1} peak was clearly apparent with a good signal-to-noise ratio; while, no significant background signal such as luminescence was observed in the measurement of a blank sample (the relevant data is not shown here). This sensitivity is difficult to compare directly with those previously reported for NF-based substrates as described in the introduction, because the tested molecules were different (dissimilar Raman cross-sections); however, the measured concentration in the present study was lower. Overall, the present results inform that Ag-Au@NF allows reproducible and sensitive SERS measurements. For reference, the response curves obtained using Sn-Ag@NF and Ag@NF are also provided in the supplementary data (Figure S5). As shown, the peak intensities in the cases using both substrates were much lower compared that employing Ag-Au@NF.

3.2. Test of physical durability and surface stability of Ag-Au@NF

The physical durability of Ag-Au@NF was initially tested by abrading it with sandpaper. For this test, Ag-Au@NF was immersed in a 100 μM R6G solution, and then the R6G-adsorbed substrate was sandwiched between a 130-g metal block and 3000 grit sandpaper. The pressure of 1.0 g mm^{-2} was applied to the substrate, and then the metal block fixed with the substrate was moved 10 cm along the surface of the sandpaper. The abraded substrate was periodically removed and used

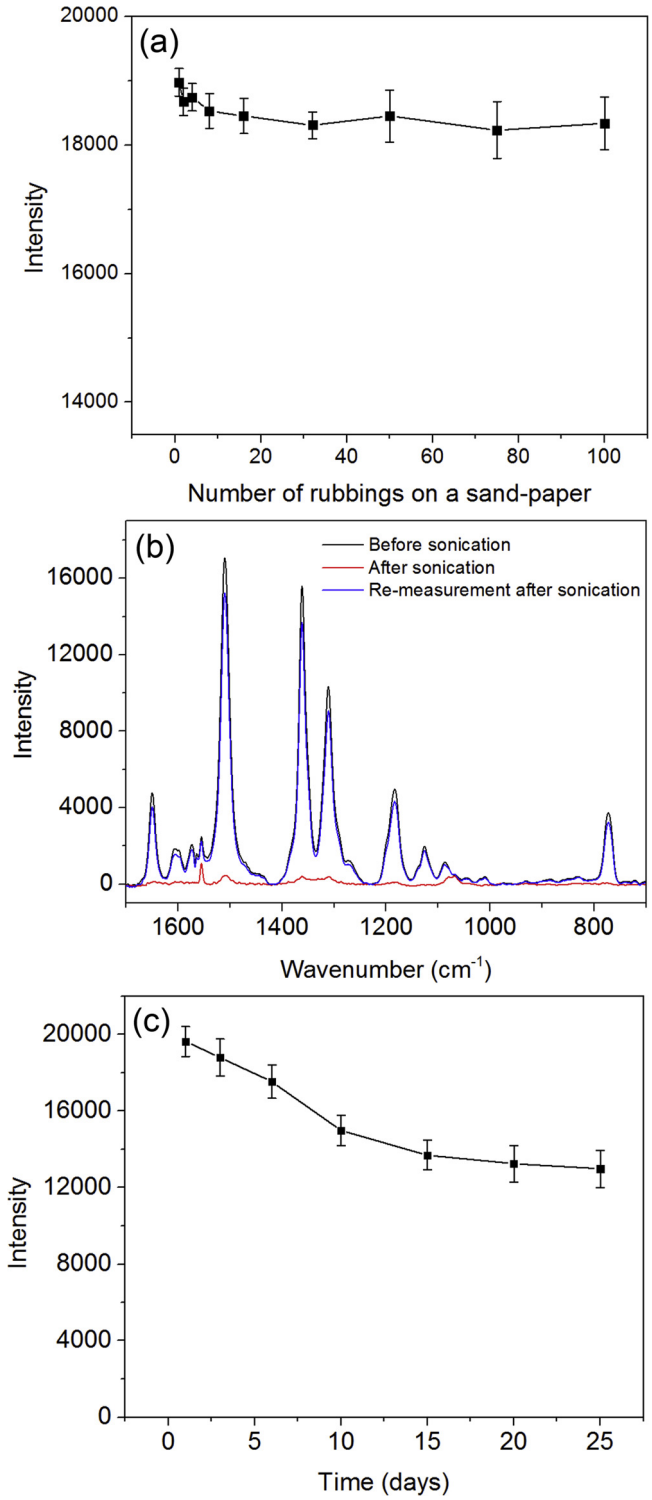

Fig. 7. Peak intensity at 1360 cm^{-1} of Ag-Au@NF with absorbed R6G during 100 cycles of sandpaper abrasion (a). Raman spectra of Ag-Au@NF with absorbed R6G, collected before and after 30 min of sonication in ethanol solution, and again after re-adsorption of R6G (b). Peak intensity at 1360 cm^{-1} of Ag-Au@NF with absorbed R6G during 25 days of exposure to ambient air (c). Error bars in (a) and (c) represent the standard deviations of 9 measurements (3 measurements on 3 different substrates).

to acquire Raman spectra; the sandpaper abrasion procedure was applied 100 times total to each of three separately prepared substrates. After the first 8 abrasion cycles, the peak intensity at 1360 cm^{-1} decreased approximately 2.35% (Fig. 7(a)), owing to removal of Ag-Au nanostructures from the exterior of the NF frame. Surprisingly, the intensity changed very little during the following 92 abrasion cycles, and was decreased by only 3.36% overall after 100 abrasion cycles. The average RSD of the measurements in Fig. 7(a) was 1.8%; while, the RSDs became slightly higher when the rubbing was repeated more than 50 times. The overall result clearly confirms that the inner Ag-Au nanostructures were well protected by the NF frame under this harsh physical treatment.

To test the physical stability of the Ag-Au nanostructures under micro-vibration, the R6G-adsorbed substrate was sonicated in an ethanol solution for 30 min. Under this sonication, the adsorbed R6G would be only detached from surface and/or Ag-Au nanostructures would be collapsed. In Raman spectra acquired from the substrate with adsorbed R6G (Fig. 7(b)), the R6 G peaks mostly disappeared after sonication, confirming the detachment of adsorbed R6 G from the surface. Next, the sonicated substrate was re-immersed in the same R6G solution and subjected to Raman analysis again. The peak intensities decreased by 12.2% relative to the initial substrate, indicating partial destruction of the nanostructures by the sonication. Nonetheless, these results represent considerable resistance to vibration. The spherical shapes of the nanostructures would be advantageous to maintain structural stability under vibration-intense environments, compared to angled or branched nanostructures.

Finally, the surface stability of Ag-Au@NF upon air exposure was tested. For this purpose, a substrate with adsorbed R6G was simply exposed to ambient conditions without storage in a sealed container, and Raman spectra of the R6G-adsorbed substrate were collected over 25 days. During the first 3 days, the intensity of the 1360 cm^{-1} peak diminished by 4.26% compared to the original intensity, and the intensity decreased by 33.8% overall during the 25 testing days test period (Fig. 7(c)). The average RSD of the measurements was 5.4%. Thus, fast surface deactivation was not observed, and the achieved stability was tolerable for routine SERS measurements. The incorporation of Au (55.9%) in the structure helped to improve the surface stability. Proper storage of the substrate in a sealed container would make the surface further stable over a longer period.

4. Conclusions

The physically durable Ag-Au@NF substrate was exceptional in abrasion and vibration tests, due to the caging of nanostructures in the sturdy NF frame and their strong adhesion, respectively. The sensitivity and surface stability of the substrate were good owing to the incorporation of both Ag and Au in the nanostructure. This substrate has balanced merits suitable for practical SERS-based analysis in field settings. Due to the rugged porous structure, introduction of sample into the Ag-Au@NF and washing of the introduced sample is very facile. Therefore, it also has potential use in online sample enrichment to enhance sensitivity. A portable analytical system integrating the developed Ag-Au@NF with a handheld Raman spectrometer is now under development and will be used for various SERS applications such as in situ detection of pesticides and antibiotics in river water.

Acknowledgements

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:https://doi.org/10.1016/j.snb.2018.11.098.

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